

Furnace Skin Temperature engineering manual

Project when the hottest thermocouple in each pass may reach 475 °C and show the temperature history behind that projection.

Run the Python model

Open furnace_skin_ti_predictor_engine.py in Python IDLE. Select Run, then Run Module (F5). Python 3.10 or newer is required; no extra packages or other files are needed.

The built-in example writes a results JSON file beside the Python file. Edit SETTINGS near the top and run again to change assumptions. To use your own measurements, run the file with an input JSON filename. Inputs use the fields and units below.

The input envelope contains schemaVersion, model, source, timezone, config and measurements. Each measurement contains timestamp, values and optional quality. Use ISO timestamps with an offset. Bad, stale and missing signals remain unavailable. The AVEVA mapping is provisional; a live connection has not been verified.

Inputs and units

Model inputs · 3 rows		
INPUT	UNIT	USE
Timestamp and each mapped skin thermocouple	date/time; °C	Current maximum and recent temperature trend for each pass.
Available pass flows and process drivers	Units defined in the supplied model bundle	Model features and the Pass 3 flow-split scenario.
Furnace/pass mapping, sampling interval and trained models	Published model bundle	Defines available thermocouples, feature names and validation results.

Calculations

TEMPERATURE TRAJECTORY

$T(d) = T_{now} + \text{recent slope} \times d$

The central projection fits temperature against actual timestamps in the latest 30 days. Missing samples retain their time gaps. It is separate from the XGBoost prediction.

UNCERTAINTY DISPLAY

Trained-horizon P10, P50 and P90 = base score + sum of tree outputs

Trees run once at the bundled training horizon. Their quantiles are reported per thermocouple, separately from the unbounded linear trend. No calibrated pass-maximum or long-range interval is claimed.

PASS MAXIMUM AND CROSSING

$T_{\text{pass}}(d) = \max(T_{\text{thermocouples}}(d))$

Interpolate the first crossing of 475 °C. A flat/falling trajectory may have no crossing. Longer extrapolations are labelled separately.

CURRENT STATUS

Alarm: $T \geq 470$ °C or crossing < 24 h; advisory: $T \geq 460$ °C or crossing < 72 h

These are review thresholds. The requested forecast target remains 475 °C.

FLOW-SPLIT SCENARIO

Pass 3 split = $100 \times Q_3 / \Sigma Q_{\text{pass}}$

Run scenario applies a simulated response to the baseline. The slider alone does not rerun the scenario.

Useful trends

- Each pass's hottest measured temperature and projected trajectory against 475 °C.
- The thermocouple driving the pass maximum and its separate trained-horizon quantiles.
- Baseline versus the applied Pass 3 scenario, with its assumptions and model-support label.

Assumptions and limits

- Missing operating inputs use the trees' missing-value branches; accuracy with incomplete inputs is unverified. Holdout accuracy describes the original training data and prediction horizon, not simulated history or the full multi-year trend extrapolation. The internal search extends up to 2,555 days and is not a validated service-life forecast.
- Original recorded history retains its 2024–2025 timestamps in Data & Log. The 2026 YTD sample replays complete operating records in reproducible blocks, interpolated to four-hour intervals. It preserves temperature/flow relationships and is labelled simulated. Block transitions are not evidence of real maintenance or process events.
- Projection starts at the last available pass observation. Its age relative to the dataset end and any missing thermocouples are disclosed. A projected crossing is not evidence of future operating conditions.
- Only mapped thermocouples contribute. An unavailable or unmapped sensor cannot establish a pass is safe.
- The flow-split response and extra scenario uncertainty are assumptions, not a plant-validated causal effect. Simulated inputs do not add validation evidence.

Measured values come from instruments or samples. Calculated values use those inputs and the configured assumptions. Design values are references. Simulated values demonstrate the model and are not plant measurements. Missing results remain unavailable.

Sources

- Supplied furnace skin-temperature workbook and published model holdout results
- XGBoost: quantile regression
- NIST: linear least squares